Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **business applications**, **web APIs**, and data workflows. Strong in **Python**, **JavaScript/TypeScript**, **SQL**, CI/CD, and reliable operational software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
M.S. in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
B.Tech. in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, OOP, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant **business platform services** on **Azure SQL** and Blob Storage for 500+ users, optimizing queries and validation flows to reduce dashboard latency from 1.2s to 350ms.
- Built fault-tolerant workflows for 1K+ background jobs with structured state transitions, failure isolation, and release-safe debugging, improving throughput 3x while supporting dependable production operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Flask** services, **REST APIs**, and instructor-facing workflows handling 10K+ daily requests, integrating secure file ingestion, rubric pipelines, and production-ready business logic for grading operations.
- Designed concurrent batch-processing for 1K+ uploaded files using multithreading and **AWS RDS**-backed services, maintaining stable API behavior while reducing turnaround time by 60% in production.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform **enterprise software** with microservices, Jenkins, PyTest, and Mockito, improving release reliability through 20+ automated test suites across Windows, Linux, and macOS.
- Optimized API and ETL workflows with **AWS Glue**, **S3**, and SQL-backed services, improving response times by 30% and pipeline efficiency by 68% for data processing.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed full-stack workflow software using **REST APIs**, WebSockets, and microservices, supporting 2M+ synchronized records while improving operational visibility, responsive interfaces, and real-time coordination.
- Automated Docker-based build and deployment workflows with **Jenkins CI/CD**, cutting release time by 50% and improving troubleshooting speed, environment consistency, and delivery reliability.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs for a cross-platform booking application, covering reservations, scheduling, status updates, and payment workflows across 50K+ records through reliable Flask services.
- Mapped 12+ integration points and implemented payment and synchronization features with **Flask**, Firebase, and webhook-driven logic, reducing manual payment handling by 70%.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
Web & Backend: Flask, FastAPI, Node.js, REST APIs, microservices, WebSockets, HTML5, CSS3, React, Next.js
Data & Databases: PostgreSQL, MySQL, Azure SQL, SQL Server, Firebase, Supabase, ETL, validation, query optimization, relational modeling
Cloud & Delivery: AWS, S3, AWS RDS, AWS Glue, Azure, GCP, Docker, Jenkins, CI/CD, Git, testing, Agile, production support

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare operations platform with scheduling, alerts, and dashboard workflows, reducing administrative workload by 30% through structured **TypeScript** services, monitoring, and business-process automation.

FinNexus Live | *Next.js, TypeScript, Plaid, MongoDB, Kubernetes*
- Built a scalable financial application supporting 50K+ users and 1M+ transactions, combining secure APIs, real-time updates, and dashboard workflows under CI/CD-backed deployment and operational monitoring.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified **Java** and TypeScript application for business data workflows.